Medical Specialty Classification from Clinical Notes: Does a Language Model Beat Classical Text Features?

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1 Introduction

The aim of this project is to build and compare natural language processing models that assign clinical texts to one of 12 medical specialties to answer the following research question:

Does a fine-tuned clinical language model that reads only a fraction of each record outperform a classical classifier that reads all of it?

To answer it, we analyse the given data (Section 2.1) and balance the classes through augmentation (Section 2.2); starting from the baseline, we develop progressively better models (Section 3), compare them on held-out data (Section 4) and study their errors (Sections 5 and 6).

2 Data

2.1 Data Analysis

Dataset. Each record, adapted from a public Kaggle dataset [Kaggle, nd], has five fields: `medical_specialty` (the label, 12 classes), `description`, `sample_name`, `transcription` and `keywords`. In `train.csv`, seven rows have a description that swallowed the records that follow; we split them back, giving 2,669 records. Descriptions are short (19 words on average) and transcriptions long (457 words; Figure 1, Appendix A).

Class distribution. Table 1 shows a strong imbalance, from Surgery with 893 records to Dermatology with 25 (about 36:1), which biases classifiers towards the frequent classes [He and Garcia, 2009].

Table 1: Records (n) per specialty in the training data (total 2,669).

Specialty	n	Specialty	n
Surgery	893	Neurology	184
Cardiovascular-Pulmonary	309	Obstetrics-Gynecology	136
Orthopedic	300	Neurosurgery	79
Radiology	222	Ophthalmology	73
General Medicine	212	Psychiatry-Psychology	41
Gastroenterology	195	Dermatology	25

Observation 1 — the same note appears under several specialties. 731 of the 1,838 distinct transcriptions are listed under more than one label (1,530 records, 57%), always with the same description and sample name. This shows that some fields are common among specialties, which can be misleading.

Observation 2 — the label is written in the keywords. In all 2,219 records that have keywords, the first keyword is the specialty itself. This could be used as a shortcut for model training, taking advantage of how the dataset was built. The other 450 records have none.

2.2 Data Augmentation

To counter the imbalance (Section 2.1), we enlarged every class below 400 records with rule-

based synthetic records. Each new record applies one of 15 variants inspired by Easy Data Augmentation (EDA) [Wei and Zou, 2019], combining synonym replacement (a clinical term map and WordNet [Miller, 1995]), word and sentence deletion, adjacent-word swaps, abbreviation expansion and number rewriting. Keywords after the first are subsampled and shuffled; negations, laterality, numbers and dosages are never altered. This added 2,979 records, raising the training file to 5,648 and reducing the ratio between the largest and the smallest class from about 36:1 to 2:1.

3 Models

We develop models step by step to beat the teachers' baseline (above 49% accuracy, no per-class F1 below 25%) while answering the research question.

M1. The baseline, the reference for every change. It reads only the description, lowercased and stemmed [Bird et al., 2009], and counts words with Term Frequency–Inverse Document Frequency (TF-IDF) for a Multinomial Naive Bayes (MNB), with default settings.

M2. Reads every field, transcription and keywords included, since the question asks what a classical model can do with all the text. Words are weighted by TF-IDF and a linear Support Vector Machine (SVM) [Pedregosa et al., 2011], learns one boundary per specialty, with class weights so that the rare ones count more. It has no extra feature and no post-processing.

M3. Goes back to little text, the description only, but swaps word counts for Bio_ClinicalBERT [Alsentzer et al., 2019], already pre-trained on clinical notes. A linear layer is added and all weights are fine-tuned with PyTorch [Paszke et al., 2019] and Transformers [Wolf et al., 2020], since the encoder was never trained to separate specialties. Training uses the augmented data.

M4. The best model: M3 plus the first keyword, which is the specialty itself (Observation 2) and, unlike the text, separates the copies of a note listed under several specialties (Observation 1). Its specialty is added to M3's standardised scores with a weight that no score gap can outvote, so the keyword decides when it exists and M3 decides when it does not. The weight is fixed in advance and M4 needs no training of its own.

4 Experimental Setup

Datasets. M1 was trained on the given `train.csv`, M2 to M4 on its augmented version (Section 2.2).

Splits. 80% of the original records are used for training and 20% (534) are held out, stratified and fixed by seed 42, so runs are comparable; augmentation is applied afterwards, to the training part only. `results.txt` comes from M4 trained on all the records; the given test file was used for nothing else.

Evaluation. Accuracy, macro-F1 and the lowest per-class F1 on the held-out records, against the 49% and 25% set by the teachers.

4.1 Parameters/Hyperparameters

Hyperparameters are set before training, unlike parameters, which are learned. Table 2 gives each model's configuration; "CV" means chosen by 5-fold cross-validation on the training part (macro-F1), "fixed" set in advance; none was tuned on the held-out records.

5 Results

Table 3 shows the held-out results; the last row applies the keyword rule of M4 to M2, for comparison. M4 is the best model (92.1% accuracy, 89.6% macro-F1) and meets both automatic conditions (accuracy above 49%; lowest class F1 70.0%,

Table 2: Configuration of each model.

M1	Description. Lowercase, Porter stemmer; TF-IDF, MNB; library defaults.
M2	Description, sample name, transcription, keywords. TF-IDF: 1–2 words (CV), terms in $\geq$2 records, at most 150,000 terms, $1 + \log tf$. Linear SVM, $C = 250$ (CV; higher = tighter fit), balanced class weights.
M3	Description, cut at 128 tokens. Bio_ClinicalBERT (release pinned in the code) + linear layer, cross-entropy. AdamW, learning rate 3×10^{-5}, weight decay 0.01 (penalty on large weights), 10% warm-up (rate grows from 0) then linear decay, gradient clipping 1.0, batch 16, 3 epochs.
M4	M3 log-probabilities standardised per record (mean 0, deviation 1), plus 5 (fixed) for the specialty of the first keyword.
All	Seed 42, Python 3.10; library versions in `code/requirements.txt`.

Psychiatry-Psychology). By label (Table 4, Appendix A), the F1 of M4 is at least 88.9% in 9 of the 12 specialties; the weakest are Psychiatry-Psychology (70.0%), General Medicine (79.5%) and Neurology (82.7%). The most frequent error (Figure 2) is Cardiovascular-Pulmonary predicted as General Medicine (6 records). Augmentation made no clear difference for M2 (87.6% original, 86.3% augmented), within the uncertainty of about 3 points of a 534-record test set.

Table 3: Held-out results (%): accuracy, macro-F1 and lowest class F1.

Model	Acc.	Macro-F1	Min. F1
M1 baseline	38.8	13.7	0.0
M2 classical, all fields	86.3	81.5	50.0
M3 language model, description	48.9	51.0	20.8
M4 = M3 + first keyword	**92.1**	**89.6**	**70.0**
M2 + first keyword (comparison)	89.1	87.2	75.0

6 Discussion

Research question. With little text, the description and the first keyword, M4 outperforms M2 (92.1% against 86.3%). However, the same keyword rule applied to M2 gives 89.1%, so the keyword explains most of the gap, and the remaining 3 points are within the uncertainty of the test set. Without the keyword, M3 reaches only 48.9%, because the description cannot separate specialties that share notes (Observation 1). The answer is yes on this dataset, but mainly because one short field carries the label (Observation 2), not because the language model reads better.

Errors. All 42 errors of M4 fall on the 94 held-out records without keywords, where M3 decides alone (55.3% accuracy); 22 of the 42 are notes listed under more than one specialty (Observation 1). Three examples (correct label $\rightarrow$ prediction): (i) “Patient felt dizzy, had some cold sweats, mild shortness of breath, . . . lost consciousness for a few seconds” (Cardiovascular-Pulmonary $\rightarrow$ General Medicine): the decisive cue, *Pulmonary Embolism*, is in the sample name, which M3 never reads. (ii) “Possible free air under the diaphragm” (General Medicine $\rightarrow$ Gastroenterology): the same note appears in the training data under Gastroenterology. (iii) “Falls at home. Anxiety and depression” (Neurology $\rightarrow$ Psychiatry-Psychology): the description is dominated by psychological symptoms.

Limitations. M4 depends on the first keyword being the specialty, which holds in all 2,219 training records that have keywords and in 320 of the 398 test records; without keywords its accuracy falls to 55.3%.

7 Future Work

With more time, we would give the language model the sample name and the start of the transcription, so that records without keywords use more text, treat the copies of a note as a multi-label problem, and use cross-validation to reduce test-set uncertainty.

Bibliography

Use of LLMs. We used Claude (Anthropic), through Claude Code, as a coding and writing assistant: it helped write and debug the scripts, run the experiments, and draft and edit the text of this report. The group defined the research question, chose the models and design decisions, and checked all results and text.

Class PDF. *"Dense vectors and pre-trained models"*

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Appendix A: Extra Figures and Tables

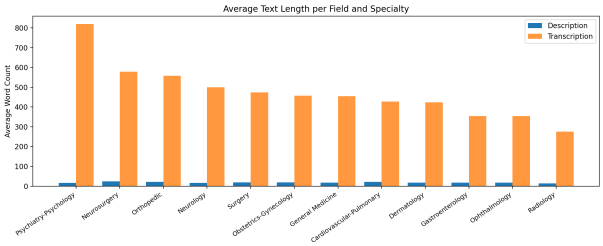


Figure 1: Average word count of the description and transcription fields per specialty.

Table 4: Results by label of M4 on the held-out records (%).

Specialty	Prec.	Recall	F1	Records
Cardiovascular-Pulmonary	94.6	85.5	89.8	62
Dermatology	100.0	80.0	88.9	5
Gastroenterology	92.5	94.9	93.7	39
General Medicine	76.1	83.3	79.5	42
Neurology	81.6	83.8	82.7	37
Neurosurgery	93.8	93.8	93.8	16
Obstetrics-Gynecology	96.3	96.3	96.3	27
Ophthalmology	93.8	100.0	96.8	15
Orthopedic	91.9	95.0	93.4	60
Psychiatry-Psychology	58.3	87.5	70.0	8
Radiology	97.5	88.6	92.9	44
Surgery	97.7	96.6	97.2	179

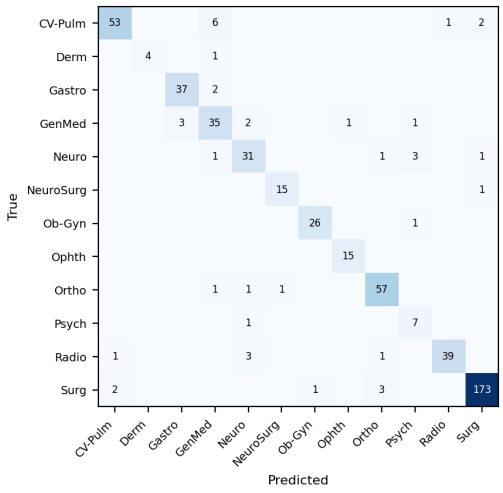


Figure 2: Confusion matrix of M4 on the held-out records (rows: true label, columns: predicted label).